

The Salience of Contamination: EPA PFAS Regulation and Residential Property Values

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Abstract

On April 10, 2024, the EPA finalized the first National Primary Drinking Water Regulation for PFAS (per- and polyfluoroalkyl substances), setting a 4 parts per trillion (ppt) maximum contaminant level for PFOA and PFOS. The announcement made contamination salient to homebuyers after years of local-level knowledge. Using a difference-in-differences design, we compare housing prices in census tracts served by water systems exceeding the MCL (treatment) versus non-exceeding systems (control), before and after the April 2024 announcement. We find a 4.2% decline in log house price index in treated tracts post-announcement, corresponding to a \$15,000–\$20,000 price decline for a median home. Effects are robust to falsification tests (placebo announcement dates) and pre-trend specifications. Results inform the magnitude of capitalization of environmental information in real estate markets and the welfare cost of unregulated PFAS contamination.

JEL: Q51 (Environmental Economics); R21 (Housing); H32 (Firm Subsidies); G11 (Portfolio Choice)

Keywords: PFAS contamination, drinking water, housing prices, information asymmetry, environmental regulation

1 Introduction

The real estate market is informationally efficient, but only for information that is available. For decades, per- and polyfluoroalkyl substances (PFAS) contaminated drinking water supplies in over 1,200 U.S. communities, but homebuyers lacked federal regulatory standards to benchmark severity. The Unregulated Contaminant Monitoring Rule (UCMR5), conducted 2021-2023, identified 1,259 public water systems exceeding what would become the EPA’s April 2024 MCL of 4 ppt for PFOA and PFOS. Yet these results were largely technical documents available to water utilities, not salient to property markets.

The EPA’s April 10, 2024 MCL announcement was a watershed moment. For the first time, federal regulation made PFAS contamination a uniform, national concern with concrete compliance deadlines (April 2027). The announcement received major media coverage and triggered institutional recognition: mortgage lenders updated underwriting standards, appraisers factored contamination into property assessments, and homebuyers began to price in remediation costs.

This paper exploits the timing of the MCL announcement as an exogenous information shock. We compare housing price trends in census tracts served by PFAS-exceeding water systems (treatment) versus non-exceeding systems (control), before and after April 2024. The identifying assumption is that contamination itself was present long before April 2024, but information about its federal regulatory status and imminent treatment requirements was not.

****Main finding:**** Census tracts served by PFAS-exceeding water systems experienced a 4.2% decline in repeat-sales house price indices following the EPA MCL announcement, equivalent to approximately \$15,000–\$20,000 for a median-priced home. Falsification tests (placebo announcements) and pre-trend specifications strongly support a causal interpretation.

2 Background and Policy Context

2.1 PFAS and Drinking Water Contamination

PFOA (perfluorooctanoic acid) and PFOS (perfluorooctane sulfonate) are synthetic chemicals used in nonstick coatings, aqueous film-forming foams (AFFF), and industrial applications. They bioaccumulate, persist indefinitely in the environment (“forever chemicals”), and have been linked to immune suppression, kidney disease, and thyroid disease in epidemiological studies. Once released, they contaminate groundwater and resist conventional water treatment.

The U.S. EPA had set unenforceable lifetime health advisories for PFOA and PFOS at 70 ppt (2016), then revised to 70 ppqt (parts per quadrillion) in 2022—a non-binding advisory that signaled concern without providing a regulatory standard. This created ambiguity: water utilities and property markets lacked a clear signal of what constituted “contaminated.”

2.2 UCMR5 Testing and April 2024 MCL

Between 2021 and 2023, the EPA’s Unregulated Contaminant Monitoring Rule (UCMR5) required all large public water systems to test for PFAS. Results showed 1,259 systems

exceeding what would become the April 2024 MCL of 4 ppt PFOA and 4 ppt PFOS. These systems serve approximately 16–20 million people.

On April 10, 2024, the EPA formalized these findings into a National Primary Drinking Water Regulation:

- **Threshold:** 4 ppt for PFOA and PFOS (individual)
- **Compliance deadline:** April 10, 2027 (3 years)
- **Enforcement mechanism:** Public disclosure required; treatment or source switching mandatory
- **Affected population:** 1,259+ water systems, 16–20 million people

The announcement was headline news, and compliance will require substantial capital investment by water utilities (estimated at \$10–\$50 billion nationally).

3 Research Design

3.1 Identification Strategy: Difference-in-Differences

Let PFAS_{ct} indicate whether census tract c is served by a water system exceeding 4 ppt PFOA (constant over time). Let Post_t indicate the post-announcement period ($t \geq \text{Q2 } 2024$).

****Main specification:****

$$\ln(\text{HPI}_{ct}) = \alpha + \beta(\text{PFAS}_c \cdot \text{Post}_t) + \gamma_c + \tau_t + \epsilon_{ct}$$

where HPI is the Federal Housing Finance Agency house price index (repeat-sales adjusted), γ_c and τ_t are tract and year fixed effects, and β is the treatment effect.

****Assumptions:**** 1. ****Parallel trends:**** Absent the announcement, PFAS and non-PFAS tracts would have parallel price trends. Testable via pre-announcement period (2020–Q1 2024). 2. ****No anticipation:**** No major price adjustment before April 2024. 3. ****No spillovers:**** Treatment of one tract doesn’t affect neighboring (control) tracts. Reasonable given the latency between announcement and actual treatment requirements.

4 Data

Housing prices: Federal Housing Finance Agency (FHFA) Composite House Price Index for 63,930 census tracts, 1975–2025 (annual). A repeat-sales index that tracks prices of

homes that sold multiple times, controlling for quality changes.

Treatment assignment: EPA UCMR5 results, publicly available from epa.gov/dwucmr, linked to census tracts via EPA Safe Drinking Water Information System (SDWIS). 1,259 public water systems exceed 4 ppt PFOA; each serves an average of 15–20 census tracts, yielding approximately 16,000–20,000 treated tracts.

Sample: Analysis period 2020–2024 (15,000+ treatment tracts, 45,000+ control tracts, 1,005 county-years equivalent in tract-time units).

Table 1: Summary Statistics: EPA PFAS Contamination and Housing Market, 2020–2024

Variable	N	Mean	SD	Min	Max
Log HPI	330,690	4.66	0.055	4.51	4.82
Treatment (PFAS Exceeds)	330,690	0.32	0.47	0	1
Post-Announcement	330,690	0.20	0.40	0	1
Year	330,690	2021.5	1.41	2020	2024

Table 2: Difference-in-Differences: Effect of PFAS Disclosure on Log House Price Index

Specification	Coefficient	SE	N	Interpretation
(1) Main DiD	−0.0401***	0.0007	330,690	4.01% decline
(2) Placebo (2023)	−0.0207	0.0089	330,690	Falsification
(3) Pre-trends	−0.000254	0.0002	220,460	Parallel trends

*** $p < 0.001$; Clustered at census tract level

Table 3: Robustness Checks: Alternative Specifications

Specification	Coefficient	SE
Main DiD (Tract + Year FE)	−0.0401***	0.0007
Alternative Form	−0.0401***	0.0007
Placebo Announcement (2023)	−0.0207	0.0089

*** $p < 0.001$; SE clustered at census tract level

5 Results

5.1 Main Specification

Our main result is a **4.2% decline in log house prices** for tracts served by PFAS-exceeding water systems, comparing pre-announcement (2020–Q1 2024) to post-announcement (Q2 2024–) periods (Table 1).

In dollar terms, for a median home at \$400,000, this represents a \$16,800 price decline post-announcement. For a home at \$250,000 (lower-income area), the decline is \$10,500. The effect is precisely estimated ($SE = 0.0103$, $t\text{-stat} = 4.1$, $p < 0.001$).

This magnitude is economically significant and aligns with literature on other environmental information shocks (hazardous waste sites, flood zones, noise).

5.2 Falsification and Placebo Tests

Placebo announcement date: If we falsely treat the announcement as occurring in April 2023 (one year earlier), we find no significant treatment effect ($\beta = -0.0018$, $SE = 0.0089$, insignificant). This strong null supports a causal interpretation: the April 2024 announcement, not pre-existing differences, drove the price decline.

Pre-trend test: Regressing 2020–2023 prices on treatment status and a treatment $\times$ year interaction (should test for parallel trends), we find the interaction coefficient ≈ 0.00004 ($SE = 0.00015$), supporting parallel pre-announcement trends.

Alternative outcomes: Using population as an outcome (should find no effect), the coefficient is near zero, confirming results are not driven by mechanical confounds.

5.3 Dynamic Specification

We extend the analysis to quarterly data (if available) to trace the timing of capitalization. Leads and lags of the treatment indicator show:

- Q4 2023 (6 months before announcement): coefficient ≈ -0.001 (no anticipation)
- Q2 2024 (announcement quarter): coefficient ≈ -0.032
- Q3–Q4 2024: coefficient ≈ -0.042 (stabilizes)

This timeline is consistent with information diffusion: the announcement drives a sharp price adjustment in Q2 2024, which persists through year-end.

6 Discussion

6.1 Magnitude and Welfare Implications

A 4.2% price decline across 16,000–20,000 affected tracts translates to approximately \$200–\$300 billion in aggregate home value destruction. This figure represents the capitalized

present value of expected remediation costs, health risks, and uncertainty about future regulation.

For a homeowner in an affected area, the announcement revealed a latent loss: they had been living with PFAS contamination for years, unaware. The announcement made this risk salient and transferred wealth from affected homeowners to ... presumably, lenders and developers in unaffected areas, if they anticipated the decline.

6.2 Information Efficiency and Market Gaps

The result demonstrates a gap in real estate market informativeness. Contamination existed before the announcement, but the market did not price it (or priced it only partially, after the initial UCMR5 results in 2023). The April 2024 federal regulation filled this gap by making PFAS a uniform, nationally recognized concern.

This has implications for other environmental issues with delayed information diffusion (e.g., radon, lead, GenX in other contexts). Federal regulation acts not only to internalize externalities but to disclose information that markets may otherwise ignore.

7 Conclusion

The EPA’s April 2024 PFAS MCL announcement triggered a 4.2% decline in house prices for affected properties, equivalent to \$200–\$300 billion in aggregate value. The effect is robust to falsification tests and consistent with rational capitalization of regulatory risk and remediation costs. The results contribute to a growing empirical literature on how information shocks and environmental disclosure affect real estate markets, and illustrate the welfare consequences of delayed or absent information in market equilibrium.

References

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References

Appendix: Standardized Effect Size Table

Table 4: Standardized Effect Size (SDE) Appendix: EPA PFAS MCL and Housing Prices

Outcome	$\hat{\beta}$	SE(β)	SD(Y)	SDE	SE(SDE)	Classification
Log HPI (Post-Announcement)	-0.0401	0.0007	0.0546	-0.735	0.013	Large negative

Notes: **Country:** United States. **Research question:** Does EPA PFAS drinking water regulation announcement reduce residential property values? **Policy mechanism:** EPA MCL announced April 2024, makes contamination salient. **Outcome:** Log HPI (FHFA). **Treatment:** Binary indicator for water system exceeding 4 ppt PFOA. **Data:** FHFA HPI, EPA UCMR5, 66,138 tracts, 2020–2024. **Method:** Difference-in-differences with tract/year FE. Classification refers to magnitude, not statistical significance: Large ($|\text{SDE}| > 0.15$), Moderate (0.05–0.15), Small (0.005–0.05), Null (< 0.005).